

Akshay Shivashankar Bharadwaj

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Summary

Software Engineer with 2.5+ years of experience building backend services, distributed systems, and cloud infrastructure at Qualcomm. MS in Computer Science at USC, **graduating December 2026**. Strong in **Python, C++, Java, and Go** with production experience across **AWS, Docker, Kubernetes, and CI/CD automation**. Seeking **New Grad /SDE I /SRE I** roles in backend and distributed systems engineering.

Education

University of Southern California, MS in Computer Science (GPA 3.86) Jan 2025 – Dec 2026
• **Courses:** Analysis and Design of Algorithms, Web Technologies, Operating Systems, Cloud and Distributed Systems

BMS College of Engineering, BE in Computer Science (GPA 4.0) Aug 2018 – Jun 2022
• **Courses:** Data Structures and Algorithms, Web and Database design, Artificial Intelligence, Machine Learning

Experience

Teaching Assistant, CS@SC Coding Camps, USC Viterbi School of Engineering - Los Angeles, CA Jun 2026 – Present
• Instructed **200+** K–12 students in **Python** programming — control flow, data structures, and debugging — taking first-time programmers to independently writing and troubleshooting their own programs.
• Taught applied **supervised learning (KNN)** and **computer vision** via hands-on Python implementation, making university-level ML concepts accessible to beginners and extending the same fundamentals to **Arduino** and drone-control labs.

Site Reliability Engineer, Qualcomm - Bengaluru, India June 2022 – Dec 2024
• Engineered a **ReactJS-Flask** internal tool integrating **ServiceNow APIs** to automate end-to-end RFC (request for change) workflows, eliminating **100+ engineering hours of manual effort per month**.
• Reduced deployment time by **50%** by building containerized microservices infrastructure using **Docker, Kubernetes (EKS), CloudFormation, and Jenkins CI/CD pipelines**, eliminating environment-parity issues across S3, RDS, Route53, VPC, and Lambda.
• Drove **cross-functional collaboration** across teams to architect secure infrastructure for **Me2U** (Qualcomm's global file transfer platform) using **VPC endpoints, DynamoDB, and S3 lifecycle policies** for thousands of internal and external users without public internet routing.
• Designed an **AWS CloudFormation** framework automating infrastructure for **70+ applications** with EC2 auto-scaling and CloudWatch/Datadog alerting; used **GitHub Copilot** for IaC authoring and Cost Explorer analysis, cutting cloud spend by **60%**.

Engineering Intern, Qualcomm - Bengaluru, India Feb 2022 – Jun 2022
• Built a **self-service AWS deployment infrastructure using CloudFormation and Lambda** that enabled teams to deploy Java applications with minimal manual input, cutting new environment setup from days to **under an hour**.

Projects

Operating Systems Kernel Development: Weenix (Unix-Like OS Kernel) Aug 2025 – Dec 2025
• Designed and implemented process and thread management in C, including scheduling, context switching, and synchronization.
• Achieved full system call support by building a **virtual file system (VFS)** within a Linux-inspired OS kernel architecture.
• Ensured robust memory isolation by implementing **virtual memory mapping** and **page fault handling** for concurrent user processes.
• Supported concurrent process creation by developing **fork()** based child process creation in the Weenix kernel.

Linearizable Distributed Key-Value Store on the Raft Consensus Protocol (C++20, gRPC, protobuf) Jan 2026 – May 2026
• Built a replicated **key-value store** on a from-scratch **Raft** implementation with leader election and log replication, sustaining **99.9% uptime**.
• Guaranteed safety and liveness under node crashes, network partitions, and delayed RPC delivery across simulated failure scenarios.
• Eliminated race conditions by designing concurrent, **thread-safe state management** for distributed RPC handlers.
• Achieved **zero data loss** and **sub-500ms leader re-election** validated via fault-injection tests under real-world failure conditions.

Artsy: Web & Mobile App using Django, PyTorch, Angular.js, Android Studio and PostgreSQL Jan 2025 – May 2025
• Delivered **sub-200ms API response times** by building an **Angular.js** frontend with a **Django REST Framework** API and GCP deployment.
• Achieved **87% accuracy** by training a **PyTorch** image classifier on artwork metadata scraped and normalized using **BeautifulSoup**.
• Built an **Android Studio** mobile client consuming the REST APIs to deliver real-time artwork discovery on mobile.

Moonquake Alert System (MAS), Space@yourService - EPFL (Swiss Federal Institute of Technology) Jul 2020 – May 2021
• Achieved **92% detection accuracy** by building a **Python-based** moonquake detection system using signal processing and anomaly detection.
• Reduced false positives by building a **multi-stage anomaly detection pipeline** using **bandpass filtering** on raw lunar rover seismic data.
• Guaranteed **astronaut safety** by containerizing the detection system on **Docker** in coordination with EPFL for the Asclepius experiment.
• Achieved **sub-5 second alert dispatch** by decoupling the detection engine from the notification layer via a **config-driven architecture**.

Technologies

Languages: Python, C++, C, Go, Java, C#, SQL, JavaScript, TypeScript, HTML, CSS, Unix Shell Scripting, Linux Kernel Programming
Frameworks & Libraries: Flask, Django, PyTorch, REST APIs, gRPC, Node.js, Express, ReactJS, AngularJS, Bootstrap, Tomcat
Tools: AWS, Docker, Kubernetes, Terraform, GCP, Jenkins, Git, Bash, Datadog, PagerDuty, MongoDB, MySQL, PostgreSQL

Certifications

AWS Certified Solutions Architect – Associate, Amazon Web Services

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